

Description of Load Testing

Load testing is a subtype of performance testing that involves collecting metrics and determining the performance and response time of a software/hardware system or device in response to an external request. This is done to establish compliance with the requirements set for that system (or device). The term "load testing" can be used with different meanings within the professional software testing community. In general, it refers to the practice of simulating expected application usage by emulating the work of multiple users simultaneously. Therefore, this type of testing is most suitable for multi-user systems, often those using a client-server architecture (e.g., web servers). However, other types of software systems can also be tested in a similar way. For example, a text or graphics editor can be forced to read a very large document, or a financial package can be made to generate a report based on data from several years. A properly designed load test yields the most accurate results.

The main goal of load testing is to create a certain expected load on the system (e.g., via virtual users) and, typically, using identical software and hardware, to observe the system's performance metrics.

Some tools for load testing are listed below:

Software	Vendor/Developer	Description
OpenSTA	Open System Testing Architecture	Free and open-source software for load/stress testing, licensed under GNU GPL. Uses a distributed application architecture based on CORBA. A Windows version is available, though it has compatibility issues with Windows Vista. Support was discontinued in 2007.
IBM Rational Performance Tester	IBM	Eclipse-based software that enables generating high-volume loads and measuring response times for client-server applications. Requires licensing.
JMeter	Apache Jakarta	Java-based cross-platform toolkit for

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	Project (Open Source)	conducting load tests using JDBC, FTP, LDAP, SOAP, JMS, POP3, HTTP, TCP connections. Allows generating a large number of requests from multiple computers and controlling the process from a single machine.
HP LoadRunner	HP	A load testing tool originally designed to emulate the work of a large number of concurrent users. Can also be used for unit or integration testing.
LoadComplete	SmartBear	Proprietary product for conducting load testing of web applications.
SilkPerformer	Micro Focus (Borland)	
Siege	Joe Dog Software	Siege is a utility for load testing web servers.
Visual Studio Team System	Microsoft	Visual Studio provides a performance testing tool, including load and unit testing.
QTest	Quotium	
HTTPerf		
QALoad	Compuware Ltd.	
(The) Grinder		